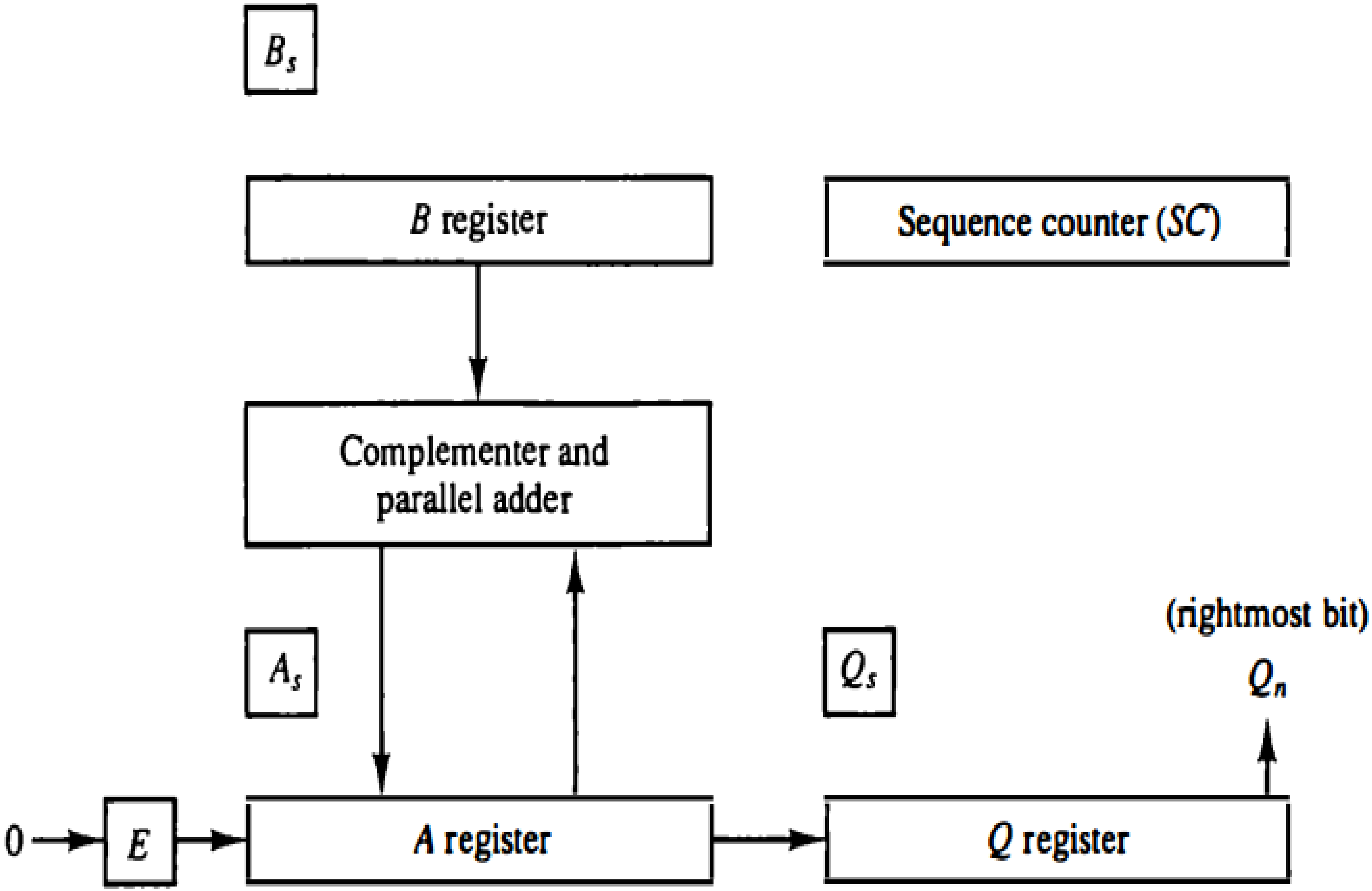
Booth's Multiplication Algorithms

- **Multiplication of two binary numbers is done by a process of successive shift and add operations**

```
 23        10111     Multiplicand
 19     x  10011     Multiplier
 --    ---------
           10111
          10111
         00000     +
        00000
       10111
       ---------
437    110110101     Product
```

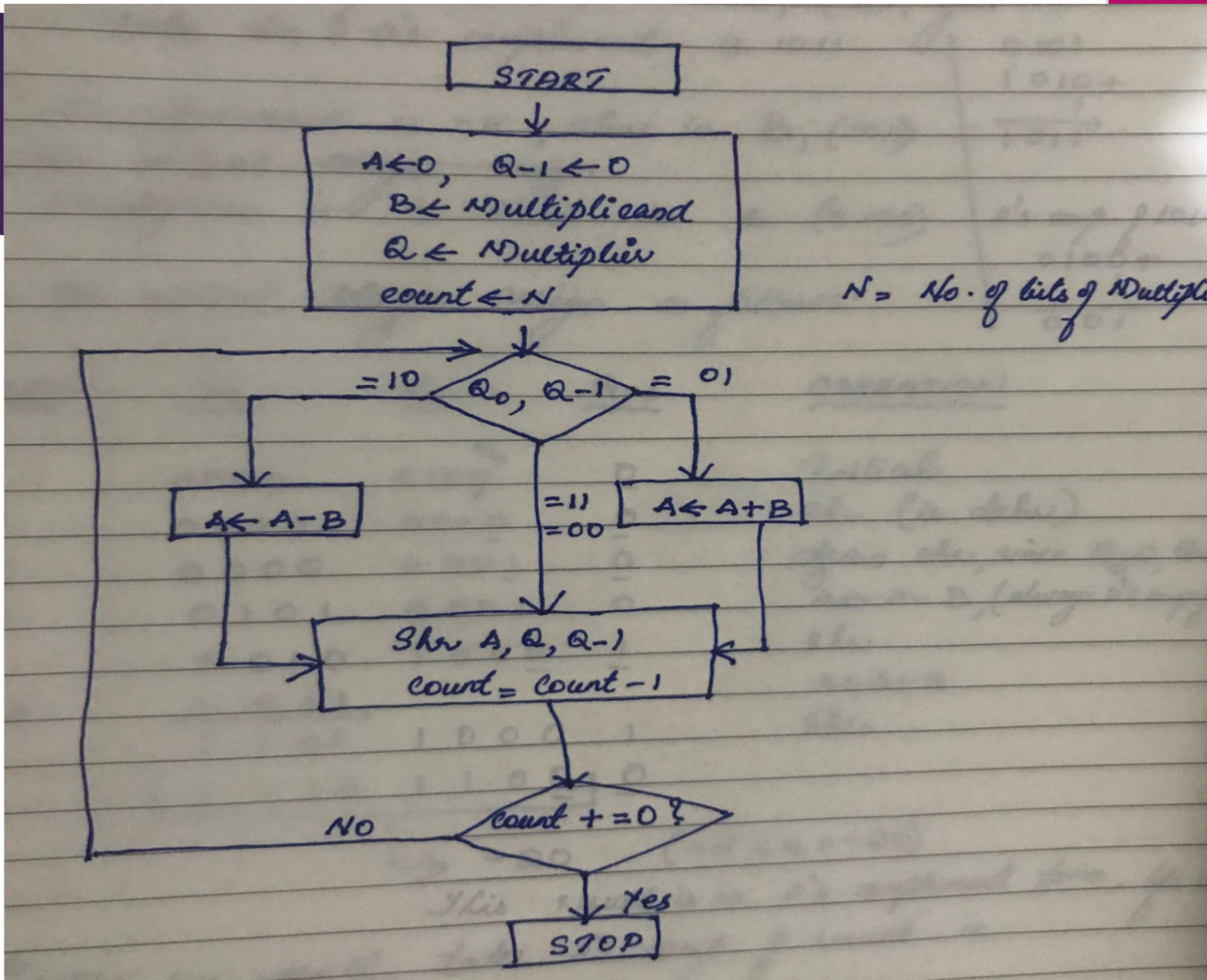
Figure 10-5 Hardware for multiply operation.



- The multiplier is stored in the Q register and its sign in Qs.
- **The sequence counter SC is initially set to a number equal to the number of bits in the multiplier.**
- The sequence counter is decremented by 1 after forming each partial product. When the content of the counter reaches zero, the product is formed and the process stops.

Booth's Algorithm for Multiplication

- Initially, the **multiplicand is in B register** and the **multiplier in Q register**. Their corresponding signs are in Bs, and Qs, respectively.



START
A ← 0, Q-1 ← 0
B ← Multiplicand
Q ← Multiplier
count ← N
N = No. of bits of Multiplier
Q0, Q-1
= 10
= 01
= 11
= 00
A ← A − B
A ← A + B
Shr A, Q, Q-1
count = count − 1
count = 0?
NO
Yes
STOP

Booth's Algorithm

Multiplication of -5×4. Here, (-5), you've to take the 2's complement. ie 1011

0101
1010+
1
1011

Multiplicand is -5, store in B, (1011)

~~In the initial stage assign~~

Multiplier is 4 store in Q (0100)

2's comp. of 1011
0100+
1
0101

In the initial stage assign as follows:

STEPS	A	Q	Q-1	OPERATION
	0000	0100 (Q_0)	0	Initial
1	0000	0010	0	Shr (ie Ashr)
2	0000	0001	0	Again shr, since $Q_0=0$, $Q_{-1}=0$
3	0101	0001	0	$A \leftarrow A-B$, (always 2's comp of B)
	0010	1000	1	Shr.
4	1011			$A \leftarrow A+B$
	1101	1000	1	Shr.
	1110	1100	0	

$\boxed{1110\ 1100} \rightarrow -20 \quad (-5 \times 4 = -20)$

This result is in 2's complement form. If u want to verify ur result, take 2's comp. of result ie

1's comp $\rightarrow$ 0001 0011+
1
0001 0100 $\rightarrow$ +20

END